

lab16(SSU)

October 27, 2025

1 Lab Assignment 16: Great British Bake Off (A/B Test)

This week's lab will focus on A/B Testing using data from the ever-popular British television show, *The Great British Bake Off*.

Helpful Resource:

- [Python Reference](#)

Recommended Readings:

- [Error Probabilities](#)
- [A/B Testing](#)

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
!pip install py pandoc
!pip install datascience
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
pandoc set to manually installed.
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynchronet2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries
  texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
  texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
  poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
  fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
```

fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 54 newly installed, 0 to remove and 38 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64 0.19-3build2 [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]

Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64 1.0.2-1build4 [45.2 kB]

Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64 2.13.1-1 [1,221 kB]

Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all
 2.004.5-6.1 [4,532 kB]
 Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all
 20201225-1build1 [397 kB]
 Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all
 20180621-3.1 [10.2 MB]
 Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java
 all 18-1 [4,720 B]
 Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-
 java all 43-1 [10.8 kB]
 Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-
 java all 1.2-2 [60.3 kB]
 Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1
 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
 Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration
 all 1.18 [5,336 B]
 Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64
 3.0.2-7ubuntu2.11 [50.1 kB]
 Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems
 all 3.3.5-2ubuntu1.2 [228 kB]
 Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
 [5,100 B]
 Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
 kB]
 Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
 0.1.1-2 [12.6 kB]
 Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
 all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
 2.5.11+ds1-1 [421 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2
 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64
 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
 1:1.0.5-0ubuntu2 [578 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
 1:7.7+6build2 [94.6 kB]
 Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
 2.004.5-6.1 [9,471 kB]

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Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 3s (57.8 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126718 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...

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Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...

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Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmldr.
Preparing to unpack .../27-ruby-xmldr_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmldr (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.

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Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...

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Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...

```



```

Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

```

```

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.
Collecting py pandoc
  Downloading py pandoc-1.15-py3-none-any.whl.metadata (16 kB)
  Downloading py pandoc-1.15-py3-none-any.whl (21 kB)
Installing collected packages: py pandoc
Successfully installed py pandoc-1.15
Collecting data science
  Downloading data science-0.18.0-py3-none-any.whl.metadata (910 bytes)
Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-
packages (from data science) (0.20.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-
packages (from data science) (75.2.0)
Requirement already satisfied: matplotlib>=3.0.0 in
/usr/local/lib/python3.12/dist-packages (from data science) (3.10.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages

```

(from datascience) (2.2.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from datascience) (1.16.2)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from datascience) (2.0.2)

Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-packages (from datascience) (7.34.0)

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (from datascience) (5.24.1)

Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages (from datascience) (0.8.2)

Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (3.1.6)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2.32.4)

Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2025.4.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (25.0)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (2.9.0.post0)

Collecting jedi>=0.16 (from ipython->datascience)

 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.4.2)

Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.7.5)

Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (5.7.1)

Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in

```

/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)
Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (2.19.2)
Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (0.2.0)
Requirement already satisfied: matplotlib-inline in
/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.1.7)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
(0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from
jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.10.5)
Downloading datascience-0.18.0-py3-none-any.whl (725 kB)
725.0/725.0 kB
13.7 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
41.7 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.0 jedi-0.19.2

```

```
[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

Mounted at /content/drive

```
[3]: # Run this cell to set up the notebook, but please don't change it.

# These lines import the Numpy and Datascience modules.
import numpy as np
from datascience import *

# These lines do some fancy plotting magic.
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plots
plots.style.use('fivethirtyeight')
import warnings
#warnings.simplefilter('ignore', (FutureWarning, np.VisibleDeprecationWarning))
```

1.1 1. A/B Testing

A/B testing is a form of hypothesis testing that allows you to make comparisons between two distributions. We may also refer to an A/B test as a permutation test.

You'll almost never be explicitly asked to perform an A/B test. Make sure you can identify situations where the test is appropriate and know how to correctly implement each step. Oftentimes, we use an A/B test to determine whether or not two samples came from the same underlying distribution.

Question 1.1. The following statements are the steps of an A/B hypothesis test presented in a *random order*:

1. Choose a test statistic (typically the difference in means between two categories)
2. Shuffle the labels of the original sample, find your simulated test statistic, and repeat many times
3. Find the value of the observed test statistic
4. Calculate the p-value based off your observed and simulated test statistics
5. Define a null and alternate model
6. Use the p-value and p-value cutoff to draw a conclusion about the null hypothesis

Assign `ab_test_order` to an array of integers that contains the correct order of an A/B test, where the first item of the array is the first step of an A/B test and the last item of the array is the last step of an A/B test.

```
[ ]: ab_test_order = make_array(5, 1, 3, 2, 4, 6)
```

Question 1.2. If the null hypothesis of an A/B test is correct, should the order of labels affect the differences in means between each group? Why do we shuffle labels in an A/B test? If you are in a lab section, confirm your answer with a neighbor or staff member before moving on.

If the null hypothesis of an A/B test is true that is, if the distributions of the two groups are identical the order of the labels should not affect the difference in the means between the two groups.

The reason labels are shuffled in an A/B test is to simulate the distribution of the test statistic that would be obtained if the null hypothesis were true. By randomly reassigning labels, we can assess whether the observed mean difference is due to chance or whether there is a real difference between the two groups. This is used to calculate the p-value by generating a distribution under the null hypothesis.

1.2 2. The Great British Bake Off

“The Great British Bake Off (often abbreviated to Bake Off or GBBO) is a British television baking competition, produced by Love Productions, in which a group of amateur bakers compete against each other in a series of rounds, attempting to impress a group of judges with their baking skills” [Wikipedia](#)

For every week of the competition, the judges assign one contestant the title “Star Baker”. Ultimately, one winner is crowned every season. Using this information, we would like to investigate how winning Star Baker awards affects the odds of winning a season of the show.

Question 2.1. We want to know whether winning more Star Baker awards *causes* a change in likelihood of winning the season. Why is it not sufficient to compare star baker rates for winners and losers?

Simply comparing the Star Baker award rates between winners and non-winners does not allow us to conclude that winning the Star Baker award causes the likelihood of winning the season. This is because correlation does not imply causation. Other factors may influence the relationship between winning the Star Baker award and winning the season, and these factors may influence both (confounding variables). For example, superior baking skills or editing effects may influence both outcomes. Asserting causality requires a more rigorous research design, such as a randomized controlled experiment, which has limitations based solely on observational data.

1.2.1 Running an Experiment

We are going to run the following hypothesis test to determine the association between winning and number of Star Baker awards. The population we are examining is every contestant from seasons 2 through 11 of GBBO. We are going to use the following null and alternative hypotheses:

Null hypothesis: The distribution of Star Baker awards between contestants who won their season and contestants who did not win their season is the same.

Alternative hypothesis: Contestants who win their season of the show will win more Star Baker awards on average.

Our alternative hypothesis is related to our suspicion that contestants who win more Star Baker awards are more skilled, so they are more likely to win the season.

Question 2.2. Should we use an A/B test to test these hypotheses? If yes, what is our “A” group and what is our “B” group?

Yes, it would be appropriate to use A/B testing to test these hypotheses.

Group A: Contestants who did not win the season (`won == 0`) Group B: Contestants who won the season (`won == 1`) We want to determine whether there is a difference in the distribution of Star Baker awards between the two groups (winners and non-winners), specifically whether winners receive more Star Baker awards on average than non-winners.

Check your answers with your neighbors or a staff member before you move on to the next section.

The `bakers` table below describes the number of star baker awards each contest won and whether or not they won their season (1 if they won, 0 if they did not win). The data was manually aggregated from Wikipedia for seasons 2-11 of the show. We randomized the order of rows as to not spoil the outcome of the show.

```
[4]: bakers = Table.read_table("./DS/star_bakers.csv")
      bakers.show(3)
```

<IPython.core.display.HTML object>

Question 2.3. Create a new table called `means` that contains the mean number of star baker awards for bakers who did not win (`won==0`) and bakers that did win (`won==1`). The table should have the column names `won` and `star baker awards mean`.

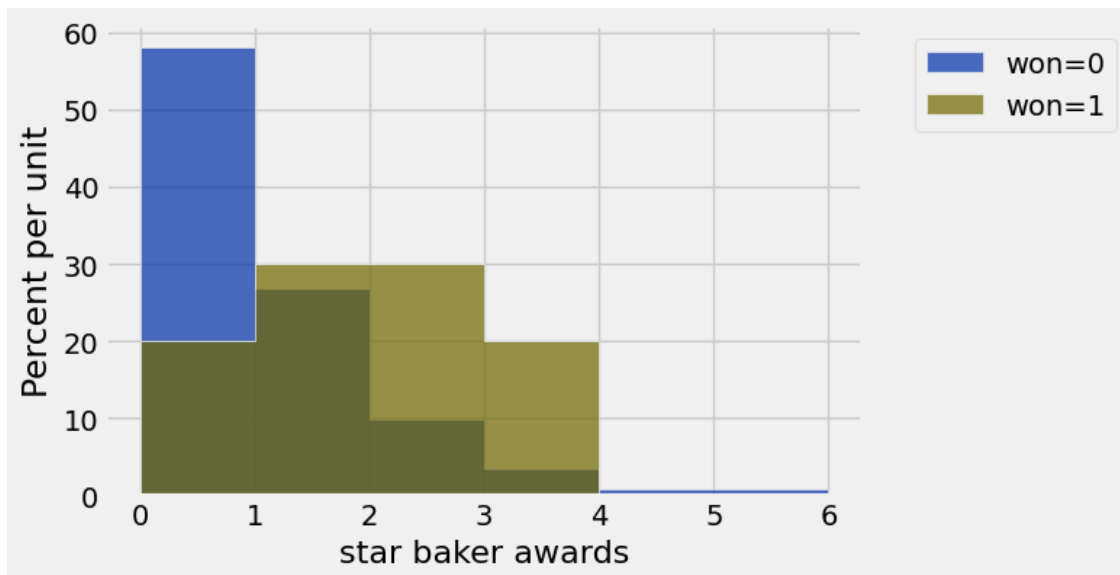
```
[5]: means = bakers.group('won', np.mean).select('won', 'star baker awards mean')
      means
```

```
[5]: won | star baker awards mean
      0 | 0.651786
      1 | 1.5
```

Question 2.4. Visualize the distribution of Star Baker awards for winners and non-winners as overlaid histograms. You should use the bins we provided.

Hint: You will want to use the `group` argument of `tbl.hist`. In order to produce several overlaid histograms based on unique values in a given column, we can do something like `tbl.hist(..., group=<col_name>, bins=...)`. This will graph one histogram for each unique value in the specified column all on a single plot.

```
[7]: useful_bins = np.arange(0, 7)
      bakers.hist('star baker awards', group='won', bins=useful_bins)
```



Question 2.5. We want to figure out if there is a difference between the distribution of Star Baker awards between winners and non winners.

What should the test statistic be? Which values of this test statistic support the null, and which values support the alternative? **Assign test_option to the number corresponding to the correct test statistic.**

1. Absolute value of the difference between the means between both groups; high values support the null
2. Absolute value of the difference between the means between both groups; low values support the null
3. Average Star Baker awards for winners - average Star Baker awards for non-winners; high values support the null
4. Average Star Baker awards for winners - average Star Baker awards for non-winners; low values support the null

Before moving on, confirm your answer with a peer or in the discussion forums.

Hint: You should think about what measures we use to describe a distribution.

```
[9]: test_option = 4
```

Question 2.6. Set `observed_difference` to the observed test statistic using the `means` table.

```
[10]: observed_difference = means.where('won', 1).column('star baker awards mean').
      ↪item(0) - means.where('won', 0).column('star baker awards mean').item(0)
      observed_difference
```

```
[10]: 0.8482142857142857
```

Question 2.7. Given a table like `bakers`, a label column `label_col`, and a values column `val_col`,

write a function that calculates the appropriate test statistic.

Hint: Make sure that you are taking the directionality of our alternative hypothesis into account.

```
[13]: def find_test_stat(tbl, label_col, val_col):
        means = tbl.group(label_col, np.mean).select(label_col, f'{val_col} mean')
        mean_winner = means.where(label_col, 1).column(f'{val_col} mean').item(0)
        mean_non_winner = means.where(label_col, 0).column(f'{val_col} mean').
        ↪item(0)
        return mean_winner - mean_non_winner

find_test_stat(bakers, "won", "star baker awards")
```

```
[13]: 0.8482142857142857
```

When we run a simulation for A/B testing, we resample by **shuffling the labels** of the original sample. If the null hypothesis is true and the star baker award distributions are the same, we expect that the difference in mean star baker awards to not change when "won" labels are changed.

Question 2.8. Write a function `simulate_and_test_statistic` to compute one trial of our A/B test. Your function should run a simulation and return a test statistic.

```
[15]: def simulate_and_test_statistic(tbl, labels_col, values_col):
        shuffled_labels = tbl.sample(with_replacement=False).column(labels_col)
        shuffled_table = tbl.drop(labels_col).with_column(labels_col,
        ↪shuffled_labels)

        return find_test_stat(shuffled_table, labels_col, values_col)

simulate_and_test_statistic(bakers, "won", "star baker awards")
```

```
[15]: -0.023214285714285743
```

Question 2.9. Simulate 5000 trials of our A/B test and store the test statistics in an array called `differences`.

```
[16]: # This cell might take a couple seconds to run
differences = make_array()

num_simulations = 5000

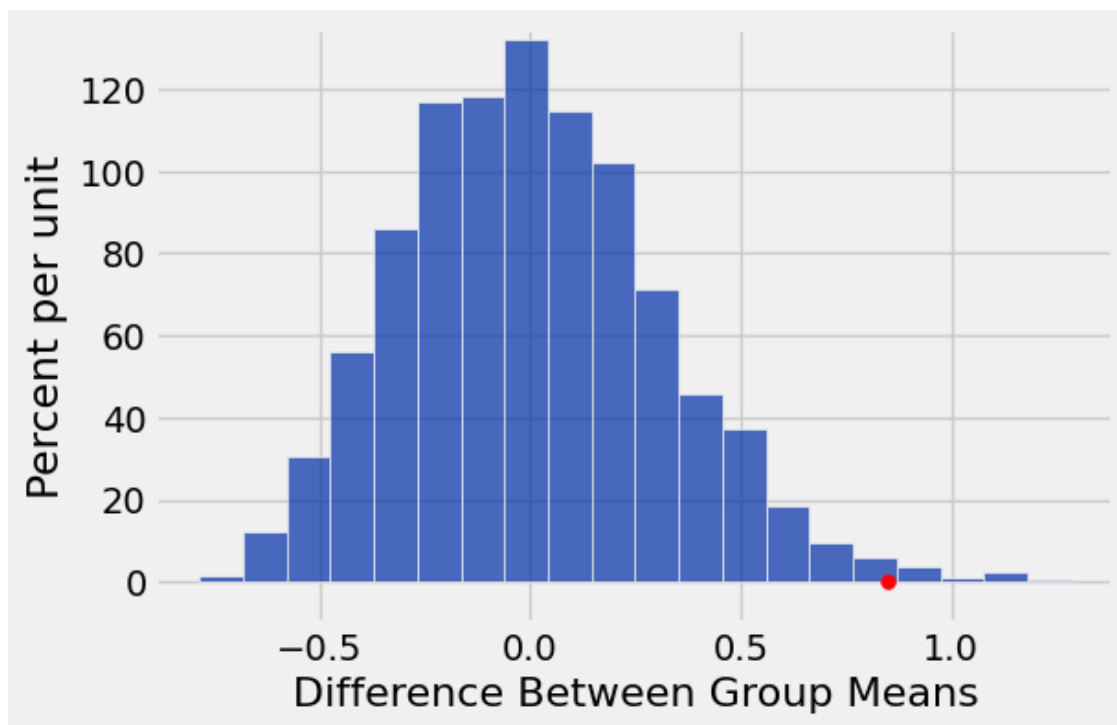
for _ in range(num_simulations):
    sim_stat = simulate_and_test_statistic(bakers, "won", "star baker awards")
    differences = np.append(differences, sim_stat)

differences
```

```
[16]: array([-0.13214286,  0.52142857, -0.13214286, ..., -0.02321429,
        -0.56785714, -0.35          ])
```

Run the cell below to view a histogram of your simulated test statistics plotted with your observed test statistic.

```
[17]: Table().with_column('Difference Between Group Means', differences).hist(bins=20)
plots.scatter(observed_difference, 0, color='red', s=30, zorder=2)
plots.ylim(-0.1, 1.35);
```



Question 2.10. Find the p-value for your test and assign it to `empirical_p`.

```
[19]: empirical_p = np.count_nonzero(differences >= observed_difference) / ↳
      num_simulations
      empirical_p
```

[19]: 0.0142

Question 2.11. Using a 5% P-value cutoff, draw a conclusion about the null and alternative hypotheses. Describe your findings using simple, non-technical language. What does your analysis tell you about the association between star baker awards and winning? What can you claim about causation from your statistical analysis? Confirm your answer with a peer, instructor or in the discussion forums.

The calculated empirical p-value is 0.0142, and the significance level (p-value cutoff) is 5% (0.05). Because the p-value (0.0142) is less than the significance level (0.05), we reject the null hypothesis.

This means there is insufficient evidence to support the null hypothesis that the distribution of Star Baker awards is identical between season winners and non-winners. Instead, we found evidence

supporting the alternative hypothesis that season winners, on average, receive more Star Baker awards.

In short, our analysis shows that contestants who win a season of The Great British Bake Off tend to receive more Star Baker awards than non-winners.

While this statistical analysis demonstrates a correlation between Star Baker awards and season wins, it cannot directly assert that Star Baker awards cause season wins. This is because, as discussed in Question 2.1, it is difficult to establish causality based on observational data alone. Contestants who win more Star Baker awards are more likely to already possess exceptional baking skills, and this exceptional skill may contribute to both winning the Star Baker award and winning the season. Therefore, our analysis demonstrates a correlation, but does not prove causation.

1.3 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit.

Please save before exporting!

```
[ ]: # should change the directory and file name matching to yours
!jupyter nbconvert './DS/lab16(SSU).ipynb' --to pdf
```